

rains that will prove tough for the bots.

Aerial delivery drones

According to a report published by McKinsey last year, drones could become an important part of the delivery supply chain. It says, “In early 2022, more than 2,000 commercial drone deliveries were occurring every day worldwide, and that number

has only grown since. While this number is still small relative to the total number of commercial deliveries, it indicates that current activity involves more than just test-flights. Drone technology has the potential to meet a range of last-mile consumer use cases, such as prepared food, convenience products, and other small packages, as well as B2B needs, such as moving medical samples to labs.”

Drones might have an edge over other delivery options in regions with poor road infrastructure. Last year, a team of researchers at the Carnegie-Mellon University also showed that using small drones for the last-mile of package deliveries uses less energy and creates fewer emissions than conventional means. Being small, fast and eco-friendly, drones do appear to be a great bet.

However, at present the labour costs involved in aerial delivery is very high. In most countries people are allowed to operate and monitor only one drone at a time. They also need to monitor the airspace in which the drone operates. This increases the labour cost involved, with each drone requiring an operator. Unless regulations allow a single operator to manage at least 20 drones, the cost of operating drones cannot even compete with other delivery options.

At present, drones are highly regulated, and many areas are out



Famous supermarket chain Carrefour has partnered with Goggo Network to trial autonomous delivery using a multi-stop capable AV (Courtesy: Carrefour)

of bounds owing to security and privacy concerns. Nevertheless, some of the industry majors are finding drone delivery to be an attractive option. Companies like Skyports and Airbus have demonstrated shore-to-ship deliveries using drones. Mattnet, another commercial drone service provider in Europe, recently launched one of the longest urban drone delivery routes to transport diagnostic samples between hospitals and laboratories in Zurich, Switzerland.

Walmart and Amazon are already delivering through drones in parts of Arizona, Arkansas, California, Florida, North Carolina, Texas, Utah, and Virginia. Walmart has 37 stores set up for drone delivery to homes and businesses. It has partnered with drone delivery service DroneUp Delivery to deliver packages that weigh up to 4.5kg. The service functions only during daytime and, according to a Walmart statement, they have more than 10,000 items available for drone delivery; the orders are delivered in around half an hour.

Amazon has started drone deliveries in parts of Northern California and Texas using its in-house Amazon Prime Air fleet of drones, which can carry items up to 2.25kg. The deliveries are monitored by Amazon employees and dropped in customers' backyards. This service too is available only in the daytime.

Sky delivery start-up Flyrtex promises five-minute delivery of food from local restaurants to customers' backyard. They too operate in parts of North Carolina and Texas in the USA.

In a January 2023 report on drones, McKinsey researchers observed that, “For drones to become truly cost-competitive across the board, operators will

need to be able to shift their focus from observing airspace to operating drones, and the number of drones per operator will need to increase significantly. That, in turn, will require technology and regulation to advance sufficiently for a single operator to manage as many as 20 drones in a densely used airspace. These advances would include autonomous drone flight in which drones fly with limited human intervention, unmanned traffic management systems, and sense-and-avoid solutions. Once those innovations are in place, regulations will need to evolve, enabling larger numbers of drones per operator. As those shifts begin to happen, the potential cost advantage of drone delivery will begin to grow.”

Self-driving trucks or vans

Self-driving trucks or vans from companies like Volvo, Caterpillar, and Komatsu have long been used off-roads for hauling goods in industries like mining, agriculture, and construction. They have also been used on roads in private environments, say, to transport goods from parts warehouses to manufacturing units within a private campus.

As the next step, now most major truck makers as well as start-ups like Aurora, Embark, Kodiak, Plus, TuSimple, Waabi, and Waymo are working on Level-4 autonomy on highways or open roads. Many